Note 4.1 Divisibility and Modular Arithmetic

Introduction

In this section, we will know the divisibility and the modular arithmetic.

Division

Definition of Divisibility

If a, b are integers and $a \neq 0$, we say that a divides b if there is an integer c such that $b = ac$ or say $c = \frac{b}{a}$. When a divides b we say a is a factor or divisor of b and that b is a multiple of a. The notation $a \mid b$ denotes that a divides b and $a \nmid b$ denotes that a does not divide b.

Theorem 1

Let a, b, c be integers, where $a \neq 0$, then

- if $a \mid b$ and $a \mid c$, then $a \mid (b + c)$.
- if $a \mid b$, then $a \mid bc$ for all integers c.
- if $a \mid b$ and $b \mid c$, then $a \mid c$.

Corollary 1

If a, b, c are integers, where $a \neq 0$, such that $a \mid b$ and $a \mid c$, then $a \mid (mb + nc)$ whenever m, n are integers.

The Division Algorithm

Theorem 2 The Division Algorithm

Let a be an integer and d a positive integer. Then there are unique integers q and r with $0 \leq r \leq d$ such that $a = dq + r$.
(Actually, it is not really an algorithm)

Definition 2

In the equality given in the division algorithm, d is called the divisor, a is called the dividend, q is called the quotient, and r is called the remainder. The notation is

$$q = a \,\mathbf{div}\, d, r = a \,\mathbf{mod}\, d$$

Both the upper two are functions on the set of integers. And we have $a \,\mathbf{div}\, d = \lfloor a/d \rfloor$ and $a \,\mathbf{mod}\, b = a - d\lfloor a/d \rfloor$.

Modular Arithmetic

Definition of Modular

If a and b are integers and m is a positive integer, then a is congruent to b modulo m if m divides $a - b$, denoted by $a \equiv b (\mathrm{mod}\ m)$. And we say $a \equiv b (\mathrm{mod}\ m)$ is a congruence and that m is its modulus (plural moduli). If a and b are not congruent modulo m, we write $a \not\equiv b (\mathrm{mod}\ m)$.

Note that here the "mod" are different.

Theorem 3

Let a and b be integers, and let m be a positive integer. Then $a \equiv b (\mathrm{mod}\ m)$ if and only if $a \,\mathbf{mod}\, m = b \,\mathbf{mod}\, m$.

Theorem 4

Let m be a positive integer. The integers a and b are congruent modulo m if and only if there is an integer k such that $a = b + km$.

Congruence Class

The set of all integers congruent to an integer a modulo m is called the congruence class of a modulo m.

Theorem 5

Let m be a positive integer. If $a \equiv b (\mathrm{mod}\ m)$ and $c \equiv d (\mathrm{mod}\ m)$, then

$$a + c \equiv b + d (\mathrm{mod}\ m), ac \equiv bd (\mathrm{mod}\ m)$$

Note that some properties seemed to be true are not valid. Such as if $ac \equiv bc (\mathrm{mod}\ m)$ then $a \equiv b (\mathrm{mod}\ m)$ may be false; if $a \equiv b (\mathrm{mod}\ m), c \equiv d (\mathrm{mod}\ m)$, then $a^c \equiv b^d (\mathrm{mod}\ m)$ may be false.

Corollary 2

Let m be a positive integer and let a and b be integers, then

$$(a + b) \,\mathbf{mod}\, m = ((a \,\mathbf{mod}\, m) + (b \,\mathbf{mod}\, m)) \,\mathbf{mod}\, m$$

$$ab \,\mathbf{mod}\, m \equiv ((a \,\mathbf{mod}\, m)(b \,\mathbf{mod}\, m)) \,\mathbf{mod}\, m$$

Arithmetic Modulo m

We can define arithmetic operations on Z_m, the set of nonnegative integers less than m, that is, the set $\{0, 1, \cdots, m-1\}$. Then we can define the $+_m$ by

$$a +_m b = (a + b) \,\mathbf{mod}\, m$$

and define the $\cdot_m$ by

$$a \cdot_m b = (a \cdot b) \,\mathbf{mod}\, m$$

And the upper are arithmetic modulo m.
It satisfies the properties:
Closure, asscosiativity, commutativity, identity elements, additive inverses($a +_m (m-a) = 0$), distributivity.